The reliability of Maze versus self-paced reading for individual differences research

Jeffrey J. Green, Brigham Young University (jjgreen@byu.edu)

A growing interest in psycholinguistics is how individual differences affect sentence processing (Cunnings, 2025; Kidd et al., 2018; Ness et al., 2023). However, methods commonly used to study group-level effects are not always suitable for individual differences research. Self-paced reading (SPR), for example, does not appear to be a reliable method for studying individual differences (Cunnings & Fujita, 2020; James et al., 2018; Jegerski, 2024). Although individual participants' overall reading times show reliable variability, the size of critical linguistic (e.g. garden path, attachment ambiguity) effects is inconsistent within individual participants. Boyce et al. (2020) provided evidence that the Maze task elicits larger effects than SPR that are more localized to the target word and less prone to spillover. **The current study investigates whether Maze is more reliable than SPR for individual differences.**

Methods— This study re-analyzed English data from Boyce et al. (2020) comparing SPR to four different versions of Maze: G-Maze (ungrammatical distractors), L-Maze (non-lexical distractors), and two types of A-Maze (distractors automatically generated using one of two different pre-trained models: Gulordava et al., 2018 or Jozefowicz et al., 2016). The dataset included RTs to three types of ambiguity: relative clause attachment as in (1), adverb attachment (2), or S vs. NP conjunction (3). The current study also compared reliability of SPR vs. A-Maze$_{\text{Gul}}$ in sentences containing semantic anomalies (4) using data from the current author's previous research.

(1) The son of the lady who politely introduced **herself/himself** was popular at the party.

(2) James will fix the car he drove **yesterday/tomorrow**, but he will need some help.

(3) The swimmer disappointed her coach**(,)** and her mother tried to console her.

(4) Sally loved to read **books/beans** from the library.

The study used the model-based approach to measure reliability (Hui & Wu, 2023). For a measure to be useful for individual differences research, there should be sufficient variation across individuals in effect size, and the effect should be reliable within participants. Variation was measured by extracting the standard deviation of the random effects from a linear mixed-effects model for RTs in the critical regions of each experiment, and reliability was determined by measuring robust within-participant split-half correlations of the random effects.

Results and discussion— Fig. 1 illustrates selected results. There was relatively high variation in random intercepts across all measures and experiments (*SD*s $\geq$ .17). Similar to previous SPR results, the random intercepts were also reliable for SPR, G-Maze, and L-Maze (*r*s $\geq$ .76). A-Maze was less consistently reliable across experiments; A-Maze$_{\text{Gul}}$ was reliable for the adverb and anomaly experiments (*r*s $\geq$ .64), but not for the relative clause and S vs. NP experiments (*r*s $<$ 0), and A-Maze$_{\text{Joz}}$ was reliable for the relative clause experiment (r = .60), but not the adverb or S vs. NP experiment (*r*s $\leq$ .10). This indicates that SPR, G-Maze, and L-Maze reliably detect individual differences in overall reading speed, but A-Maze does so only inconsistently.

The L-Maze adverb and S vs. NP experiments and the A-Maze$_{\text{Gul}}$ adverb and anomaly experiments showed significant positive split-half correlations for random slopes (Table 1). However, random slope *SD*s for the L-Maze experiments were low, suggesting that although this measure may be reliable within individuals, there is little variation in effect size. A-Maze$_{\text{Gul}}$ had larger *SD*s in random effects, especially in the anomaly experiment. Interestingly, SPR also showed a reliable split-half correlation in the anomaly experiment, although the *SD* of random slopes for SPR was lower than that of A-Maze$_{\text{Gul}}$.

Conclusion— The current results confirm the finding that SPR is not reliable for individual differences in syntactic processing. SPR may, however, reliably reflect individual differences in lexico-semantic processing, but it still has limited variance across individuals. Maze does not appear to provide a consistently reliable alternative for individual differences research. Some varieties of Maze elicited higher variance across individuals and high split-half reliability than SPR, but this was true only for some effects.

Figure 1: Variation in random effects

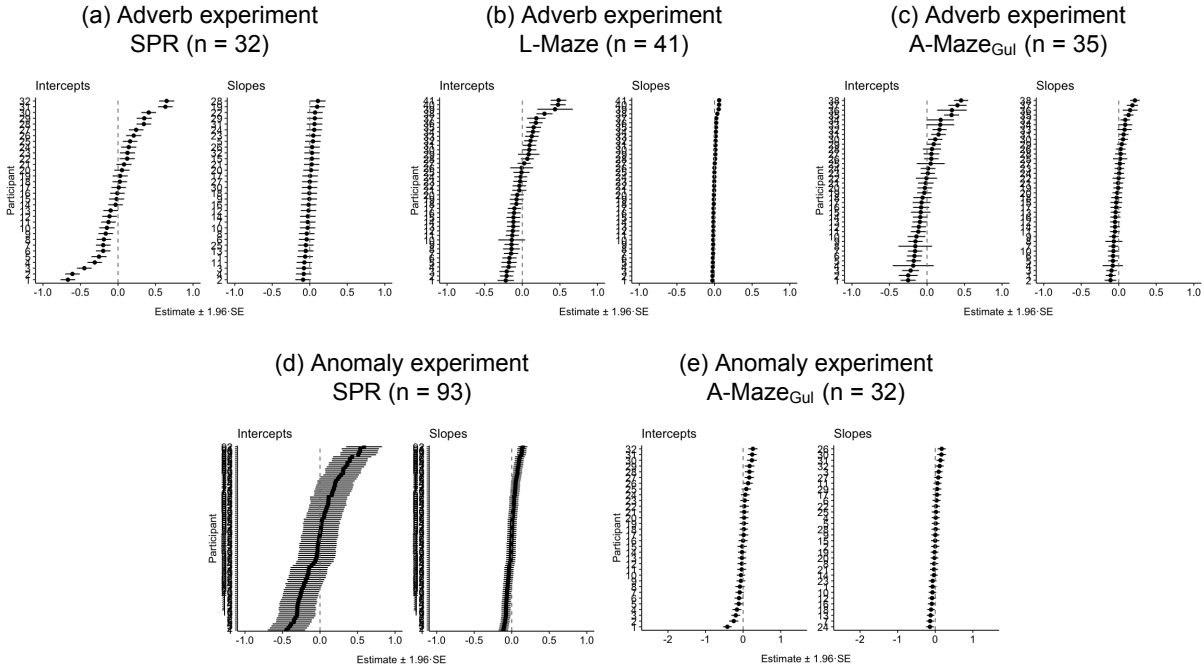

Table 1: Standard deviations and robust split-half correlations for random slopes

	SD	*r*		*SD*	*r*
Relative clauses			**S vs. NP**		
SPR	.09	.06	SPR	.05	-.34
G-maze	.08	-.20	G-maze	.02	-.84
L-maze	<.01	-.76	L-maze	.04	.81***
A-maze (Gulordava)	.07	-.58	A-maze (Gulordava)	.07	-.77
A-maze (Jocefowicz)	.01	-.24	A-maze (Jocefowicz)	.05	-.69
Adverbs			**Semantic anomalies**		
SPR	.07	.26	SPR	.07	.66***
G-maze	.08	-.20	A-maze (Gulordava)	.14	.55**
L-maze	.02	.85***			
A-maze (Gulordava)	.09	.44**			
A-maze (Jocefowicz)	.17	-.70			

Significant positive correlations: **$p < .01$, ***$p < .001$

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